PCI AI · FORMULA SHEET

Project Finance Formulas

The 16 that decide whether a deal is bankable

NPV · IRR · DSCR · LLCR · PLCR · WACC

PROJECT CONTROLS INSTITUTE GLOBAL

First edition · 2026 · AI proposes. The professional disposes.

PFL-AI · HOW TO READ THIS

Most coverage-ratio errors are not formula errors

They are definition errors and period errors.

The expression is right. The cash flow feeding it was built on the wrong basis, or discounted at a rate that does not match its own period.

This deck gives the formula, and the mistake it attracts.

The consistency rule. Discount rate, cash-flow period and compounding frequency must agree. If debt service is semi-annual, the rate is semi-annual and `n` counts half-years.

TIME VALUE · DOMAIN 3

The three that price everything

$$DF = 1 \div (1 + r)^n$$
$$AF = [1 - (1 + r)^{-n}] \div r$$
$$PV(x) = FV \div (1 + r)^n$$

A free check. The annuity factor must equal the sum of the individual discount factors over the same periods. If the two disagree, your period convention is wrong.

INVESTMENT APPRAISAL · DOMAIN 4

Appraisal, and its traps

$$NPV = \Sigma CF_t \div (1 + r)^t$$

IRR: $NPV = 0$

$$PI = PV(\text{inflows}) \div \text{Initial investment}$$

$$PI = 1 + NPV \div \text{Initial investment}$$

The last two are the same thing. Compute both. If they disagree, something in the cash-flow signs is wrong.

IRR PATHOLOGIES

Three ways IRR misleads you

Multiple IRRs. More than one sign change in the cash flows means more than one root. Check the sign pattern before quoting a number.

No IRR at all. If the flows never cross zero, there is nothing to solve.

Reinvestment blindness. IRR implicitly assumes you reinvest at the IRR itself.

> **MIRR = (FV inflows at reinvest rate ÷ −PV outflows at finance rate)^(1/n) − 1**

On mutually exclusive projects, NPV governs. IRR is blind to scale, and the highest IRR is regularly the smaller prize.

CAPITAL STRUCTURE · DOMAIN 9

Gearing is not debt-to-equity

$$\text{WACC} = (E \div V) \times Re + (D \div V) \times Rd \times (1 - t)$$
$$\text{Gearing} = D \div (D + E)$$
$$\text{Debt : equity} = D \div E$$

60% gearing is 1.5 : 1. Not 60 : 40 as a ratio. And **(1 − t)** attaches to the cost of **debt** only — the tax shield never touches the equity term.

THE WATERFALL · DOMAIN 6

The order the money moves

	PAID IN THIS ORDER
1	Operating expenses
2	Tax
3	Senior debt interest
4	Senior debt principal
5	Reserve top-ups — DSRA, MRA
6	Subordinated debt
7	Distributions to equity

Each level is paid only after the one above it.

CFADS IS PRE-FINANCING

The most consequential error in project finance

$$\text{CFADS} = \text{EBITDA} - \text{Tax paid} - \Delta\ \text{Working capital} - \text{Maintenance capex}$$

Notice what is **not** deducted. Interest.

Interest and principal are what CFADS is measured against.

Deduct interest before computing CFADS and you have double-counted it — inflating every single cover ratio in the model, in the direction that makes the deal look financeable.

DSCR · DOMAIN 10

Cover, in one period

$$\text{DSCR} = \text{CFADS} \div \text{DS}$$
$$\text{DS} = \text{Interest} + \text{Scheduled principal}$$

The minimum DSCR is the covenant. The average is informative and is never the test — a comfortable average conceals the single period that breaches.

LLCR AND PLCR

Cover, over a lifetime

LLCR = [PV(CFADS over remaining loan life) + DSRA] ÷ Debt outstanding
PLCR = [PV(CFADS over remaining project life) + DSRA] ÷ Debt outstanding

Same debt balance. Different windows.

THE TAIL

And the gap between them has a name

Tail = Project life − Loan life

PLCR should always exceed LLCR where a tail exists, because it discounts a longer stream of cash against the same debt balance.

The gap between the two ratios is the tail — the project's cash-generating life after the debt matures, and the lender's room to reschedule. A thin tail is a credit concern even when LLCR looks comfortable.

DEBT SIZING · DOMAIN 10

Sizing the debt backwards

Debt service capacity_t = CFADS_t ÷ Target DSCR
Debt capacity = Σ [Capacity_t ÷ (1 + kd)^t]

Check it by reversing. Service exactly that capacity and the DSCR lands on the target in every single period. If it does not, the discounting and the capacity are on different period conventions.

REPAYMENT SHAPE

Sculpted or annuity

Sculpted: principal_t = Capacity_t − Interest_t

Sculpted sets debt service as a constant multiple of CFADS, so DSCR is flat at the target every period.

Annuity sets a constant instalment, so DSCR rides up and down with CFADS.

Sculpting maximises debt capacity for a given minimum DSCR. That is why project finance uses it and corporate lending largely does not.

RETURNS

Two IRRs, two different questions

Project IRR — is the **asset** worth building?

Equity IRR — is the **deal** worth doing?

MOIC = Total distributions ÷ Total equity invested

A strong equity IRR on a weak project IRR is a statement about leverage, not about the asset. Quote both, or you are quoting neither.

CONSTRUCTION · DOMAIN 14

The funding requirement is bigger than the contract

IDC = Σ Interest on the drawn balance during construction
Total project cost = Construction + IDC + Fees + Working capital + Reserves

The bridge to project controls. A lender's cost-to-complete test and a controls team's EAC answer the same question from two sides. If controls forecasts an outturn above the funded amount, funding adequacy has already failed — usually before anyone in finance has been told.

MOST MISAPPLIED · 1–5

The ten that cost the most

1. Interest deducted before **CFADS** — every ratio inflated
2. Annual rate applied to semi-annual debt service
3. **LLCR** and **PLCR** treated as interchangeable
4. Average DSCR quoted as the covenant
5. **IRR** ranked above **NPV** on mutually exclusive projects

MOST MISAPPLIED · 6–10

And the other five

1. Multiple IRRs unnoticed — more than one sign change, more than one root
2. Gearing confused with debt-to-equity
3. Tax shield applied to the equity term of **WACC**
4. **IDC** and fees omitted from total project cost
5. Equity IRR presented as project performance

Each of these survives model review, because the spreadsheet computes cleanly and the output looks plausible.

The full sheet is free

This deck is 16 of them. The complete reference carries the whole quantitative surface of project finance — time value, appraisal, cost of capital, the cash waterfall, coverage ratios, debt sizing and sculpting, equity returns, construction drawdown and sensitivity — each with the mistake it attracts and reproducible worked examples.

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